

Review article

Measuring the quality of generative AI systems: Mapping metrics to quality characteristics — Snowballing literature review

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ABSTRACT

Context Generative Artificial Intelligence (GenAI) and the use of Large Language Models (LLMs) have revolutionized tasks that previously required significant human effort, which has attracted considerable interest from industry stakeholders. This growing interest has accelerated the integration of AI models into various industrial applications. However, the model integration introduces challenges to product quality, as conventional quality measuring methods may fail to assess GenAI systems. Consequently, evaluation techniques for GenAI systems need to be adapted and refined. Examining the current state and applicability of evaluation techniques for the GenAI system outputs is essential.

Objective This study aims to explore the current metrics, methods, and processes for assessing the outputs of GenAI systems and the potential of risky outputs.

Method We performed a snowballing literature review to identify metrics, evaluation methods, and evaluation processes from 43 selected papers.

Results We identified 28 metrics and mapped these metrics to four quality characteristics defined by the ISO/IEC 25023 standard for software systems. Additionally, we discovered three types of evaluation methods to measure the quality of system outputs and a three-step process to assess faulty system outputs. Based on these insights, we suggested a five-step framework for measuring system quality while utilizing GenAI models.

Conclusion Our findings present a mapping that visualizes candidate metrics to be selected for measuring quality characteristics of GenAI systems, accompanied by step-by-step processes to assist practitioners in conducting quality assessments.

Contents

1.	Introduction	2
2.	Background and related work	3
2.1.	ISO/IEC 25023 standard for system quality	3
2.2.	Key definitions	3
2.3.	Related work on GenAI system quality evaluation	3
3.	Research methodology	3
3.1.	Research planning	3
3.1.1.	Research objective	3
3.1.2.	Research questions	3
3.2.	Snowballing start-set	4
3.3.	Snowballing forward and backward	4
3.3.1.	Inclusion and exclusion criteria	4
3.3.2.	Rigor and industry relevance evaluation	4
3.4.	Data collection and analysis	6

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3.4.1.	Steps for mapping metrics to quality characteristics	6
3.5.	Steps for identifying evaluation methods	8
3.6.	Synthesizing processes for risky output evaluation	8
4.	Results	8
4.1.	RQ1: What metrics can be used to evaluate the quality characteristics defined by ISO/IEC 25023 for generative AI system outputs?	8
4.1.1.	Mapping metrics to ISO quality characteristics	9
4.2.	RQ2: What methods or approaches exist that could utilize the identified metrics to assess the quality of outputs produced by generative AI systems?	11
4.2.1.	Human evaluation method	11
4.2.2.	Automated evaluation method	12
4.2.3.	AI-assisted evaluation method	14
4.2.4.	Summary of human, automated, and AI-assisted evaluation	14
4.3.	RQ3: What processes can be used to apply metrics for evaluating the potential of risky outputs generated by generative AI systems?	15
5.	A suggested framework for evaluating generative AI systems	17
6.	Validity threats	18
6.1.	Study selection validity	18
6.2.	Data validity	18
6.3.	Research validity	18
7.	Discussions	18
7.1.	Datasets required for automated metrics	18
7.2.	Cost versus quality	18
7.3.	Selecting appropriate metrics to evaluate system output quality	18
7.4.	Risky outputs exist in GenAI industrial applications	18
7.5.	The potential for undiscovered risky outputs and the need for new metrics	19
7.6.	Limitations of snowballing with a small start-set	19
8.	Conclusions	19
	CRediT authorship contribution statement	20
	Declaration of competing interest	20
	Acknowledgment	20
	Appendix. Selected papers after snowballing literature review	20
	Data availability	20
	References	20

1. Introduction

Generative Artificial Intelligence (GenAI) [1] has emerged as a transformative technology across various fields, applied for a variety of tasks ranging from natural language processing to artifact generation [2]. GenAI systems refer to software that can generate new content using AI models based on learned patterns from training data. The content includes text, code, images, audio, and video. These AI models cover large language models (LLMs), diffusion models, and generative adversarial networks [3]. However, in this study, when we reference GenAI we explicitly mean systems based on LLMs [4].

The principal idea of GenAI systems is to utilize the capabilities of AI models to produce, manipulate, or analyze content in ways that mimic human-like behavior or understanding [2]. The integration of AI models in software products is increasingly deployed across diverse industrial domains. Key application areas include document generation [5], data analysis and insights [6], marketing and content creation [7], customer service chatbots [8], multilingual business communication [7], and code generation [9]. These domains represent critical use cases where GenAI outputs must meet specific quality standards.

However, existing evaluation methods often lack alignment with industrial quality requirements [10]. Notably, key risks include the quality and consistency of GenAI outputs across different system releases [11–13]. These risks arise from GenAI's inherent characteristics, such as non-deterministic behavior — where identical inputs may produce varying outputs, and the tendency to hallucinate [14,15] — generating content not based on factual data. Such characteristics complicate efforts to ensure reliable and trustworthy outputs in GenAI applications.

Given these risks, existing studies [16,17] suggest the use of metrics to measure the quality of AI-generated outputs. It is suggested that metrics are valuable for organizations to track and manage whether AI-generated outputs meet industry-specific quality requirements. Metrics

in this study refer to quantifiable measures that assess the quality of the outputs produced by systems that incorporate AI rather than evaluating the underlying AI models themselves [18]. Examples of such metrics, such as BERTScore [19], COMET [20], and MoverScore [21], have been developed to assess aspects like correctness, accuracy, and relevance of GenAI system outputs.

Despite the availability of various metrics, existing literature lacks a standardized approach for categorizing these metrics in relation to quality characteristics specific to GenAI systems. Moreover, many metrics are introduced without clearly defining the quality aspects they measure, often focusing on terms like correctness, accuracy, or relevance without providing explicit mappings to recognized quality characteristics.

As such, this study examines existing metrics and methods used for evaluating GenAI system outputs. Through a snowballing literature review, we identify existing evaluation metrics and map them to quality characteristics defined by the ISO/IEC 25023 standard [22]. Details of the mapping are present in Section 3.4.1. This standard was chosen for the mapping due to its widespread recognition and applicability across various industries, offering a standardized framework for evaluating GenAI systems. Furthermore, we analyze how these metrics can be employed to assess risks associated with GenAI system outputs, such as inaccuracies or biases.

The main contributions of this study are:

- Mapping metrics to quality characteristics of generative AI systems.
- Evaluation methods showing metrics' applicability to assess the quality of outputs produced by generative AI systems.
- General processes for using metrics to assess risky outputs produced by generative AI systems.
- A descriptive framework explaining how to select evaluation methods to assess specific quality characteristics based on the GenAI system's requirements.

The rest of this paper is structured as follows: Section 2 introduces related work. Section 3 illustrates the snowballing literature review methodology. Section 4 presents the results of literature review based on extracted data. Section 5 introduces a suggested framework synthesized from the study results. Section 6, we discuss threats to the study validity. Section 7 discusses the findings and implications of our research. The conclusions and ideas for future work are in Section 8.

2. Background and related work

This section presents background information and study's related work.

2.1. ISO/IEC 25023 standard for system quality

ISO/IEC 25023 is a standard defining eight measurable quality characteristics for software systems: Functional Suitability, Reliability, Usability, Performance Efficiency, Maintainability, Compatibility, Portability, and Security. Unlike task-specific metrics, this standard provides an approach to evaluating system-level quality, which is critical for GenAI applications deployed in industrial settings. For example, Usability ensures outputs align with user needs (e.g., output accuracy), while Security addresses risks like sensitive data leakage in generated content. Our work adopts ISO/IEC 25023 due to its industry-wide applicability and emphasis on standardized quality evaluation.

2.2. Key definitions

In this paper, a 'metric' and a 'quality characteristic' are defined as follows:

- *Metric*: a quantifiable measure used to assess how well a system performs in relation to certain quality characteristics. It translates abstract quality aspects (e.g., usability, reliability) into concrete and measurable values to evaluate the system against predefined quality characteristics defined in standards (e.g., ISO/IEC 25023).
- *Quality characteristic*: a property of a software system or product that can be measured to determine its quality. We adopt the terminology and structure defined by ISO/IEC 25023 [22], where 'quality characteristic' refers to system quality aspects [16].
- *Risky output* is defined as output produced by GenAI systems that contain errors due to factors such as uncertainty propagation, non-replicability, contextual misalignment, logical inconsistencies, or data hallucinations [23].

2.3. Related work on GenAI system quality evaluation

Industrial applications that use GenAI models [24,25] include text generation [21,26,27], document summarization [6,28,29], and code generation [9,30,31]. However, studies evaluating these applications have identified several challenges in LLM outputs, such as (1) accuracy [8], (2) relevance [9], (3) coherence [32], and (4) consistency [33]. These challenges underscore the need for quality evaluation methods for GenAI systems in industrial applications.

Traditional manual evaluation methods [34] rely on individuals with domain expertise to assess the quality of AI-generated content. These approaches are often resource-intensive and prone to biases and subjective judgments from evaluators [35]. Due to these limitations, automated evaluation methods [36] become more popular in many cases. While these methods improve the time efficiency and scalability of the evaluation, they also introduce a steep learning curve for evaluators due to system setup complexity.

Prior studies, such as [14,37–40], have developed metrics for specific GenAI evaluation in a few domains, such as document generation (BERTScore [19], METEOR [41]), customer service chatbots (RUBER [42]), and code generation (PASS@k [31]). However, these metrics are often siloed by domain and lack alignment with standardized

quality characteristics. For instance, while FActScore [26] evaluates factual accuracy (Reliability), no metrics explicitly address Security (e.g., detecting sensitive data in outputs). This fragmentation limits organizations' ability to evaluate GenAI systems. Our work mitigates this gap by identifying a detailed inventory of metrics and practical methods for evaluating GenAI system qualities.

3. Research methodology

We conducted a snowballing literature review [43,44] combining peer-reviewed and grey literature to account for the rapid evolution of GenAI.

As illustrated in Fig. 1, our research process is structured into four key stages: Planning, Snowballing start-set, Snowballing forward & backward, and Data collection & analysis. The process began with defining research questions, followed by database searches and filtering to establish an initial set of studies. A snowballing method [43] was then adopted to expand the study set through forward and backward searching. The searching was iterated until no new papers were found. Based on the searching results, inclusion and exclusion criteria were applied. Furthermore, Ivarsson and Gorschek's Rigor/Relevance filtering [45] was executed to select papers based on industrial relevance criteria. Finally, data extraction and synthesis were conducted to derive insights that address the research questions.

3.1. Research planning

This section introduces our research objective and research questions.

3.1.1. Research objective

This study aims to identify existing metrics for evaluating the outputs of GenAI systems and map these metrics to the quality characteristics defined in ISO/IEC 25023, offering guidance on selecting metrics and approaches for assessing GenAI system quality.

3.1.2. Research questions

To achieve our research objective, four research questions have been formulated:

1. RQ1: What metrics can be used to evaluate the quality characteristics defined by ISO/IEC 25023 for generative AI system outputs?
RQ1 seeks to identify existing metrics that are applicable for assessing the quality of generative AI system outputs and map these metrics to quality characteristics to provide a finer-grained view of quality. The mapping can either refine or complement the existing measures predefined by ISO/IEC 25023. This study provides an overview of the metrics and their mapping to quality characteristics but does not include detailed formulas for each metric.
2. RQ2: What methods or approaches exist that could utilize the identified metrics to assess the quality of outputs produced by generative AI systems?
RQ2 targets the existing evaluation methods that use metrics to assess the quality of outputs from generative AI systems. Answers for RQ2 could be useful for AI developers, researchers, and quality assurance professionals to select methods for ensuring that the outputs generated by GenAI systems meet their quality requirements.
3. RQ3: What processes can be used to apply metrics for evaluating the potential of risky outputs generated by generative AI systems?
RQ3 aims to identify the processes for using metrics to evaluate the risky outputs generated by GenAI systems. Addressing inaccurate or harmful outputs is important for risk management teams. By establishing clear processes for using metrics, organizations can enhance the transparency and trustworthiness of GenAI systems.

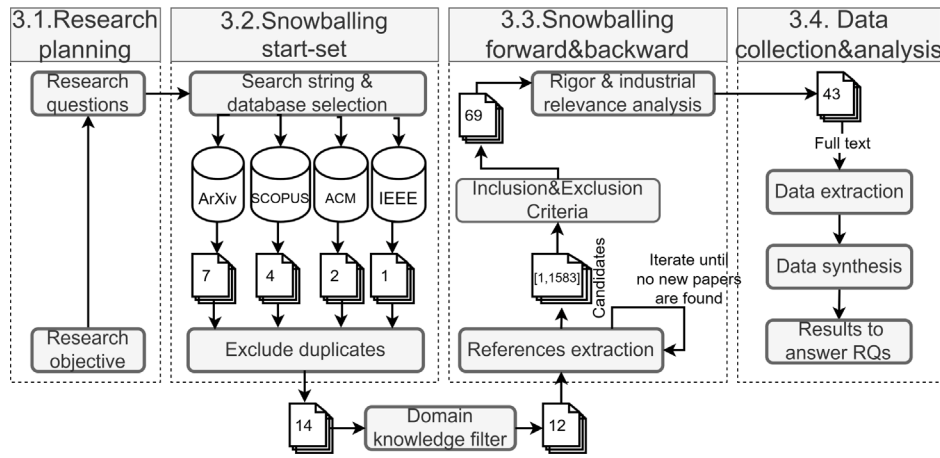


Fig. 1. Research process overview.

Table 1

Snowballing start-set literature: grey literature on Arxiv was included. Note: the citation was collected from Google Scholar on 27th April 2024.

ID	Paper title	Year	Citation
P1	A Survey on Evaluation of Large Language Models	2024	497
P2	A Survey of Evaluation Metrics Used for NLG Systems	2023	198
P3	Evaluating large language models: A comprehensive survey	2023	67
P4	A Survey on Large Language Models Applications, Challenges, Limitations, and Practical Usage	2023	63
P5	Survey on factuality in large language models Knowledge, retrieval and domain-specificity	2023	61
P6	Trends in Integration of Knowledge and Large Language Models: A Survey and Taxonomy of Methods, Benchmarks, and Applications	2023	34
P7	Large Language Models for Forecasting and Anomaly Detection A Systematic Literature Review	2024	22
P8	A Review on Large Language Models: Architectures, Applications, Taxonomies, Open Issues and Challenges	2024	16
P9	A Survey on Large Language Models for Software Engineering	2023	8
P10	Robustness, Security, Privacy, Explainability, Efficiency, and Usability of Large Language Models for Code	2024	4
P11	Machine learning for requirements engineering (ML4RE) A systematic literature review	2024	1
P12	Large Language Models for Blockchain Security A Systematic Literature Review	2024	1

3.2. Snowballing start-set

Following the guidelines set by Wohlin for snowballing [44], we initiated our literature search by identifying a start-set of recent surveys or literature review studies. Based on the start-set, we conducted backward snowballing to capture prior studies referenced in recent literature and forward snowballing [46] to identify newer studies that might not be cited in the existing literature. This combined approach ensures the coverage in a rapidly evolving field.

Firstly, we created and refined keywords to formulate search strings through pilot testing. Since the study focuses on existing metrics with industrial applicability, in addition to covering metrics for known generative AI-based technology, the search string also included an emphasis on papers that report empirical or industrial results. The final search string is shown below:

```
((TITLE-ABS-KEY("large language model") OR TITLE-ABS-KEY(LLM)) AND (TITLE-ABS-KEY(metric*) AND (TITLE-ABS-KEY("systematic literature review" OR survey) AND TITLE-ABS-KEY(experience OR empirical OR industry*)))
```

Secondly, we selected IEEE Xplore and ACM Digital Library as bibliographic databases, alongside Scopus and ArXiv. While Scopus, IEEE Xplore, and ACM Digital Library primarily index peer-reviewed publications, ArXiv was included due to its relevance in capturing recent research field of GenAI.

These databases were chosen for their coverage of computer science and engineering research, inclusion of peer-reviewed publications, and ability to capture recent developments in the rapidly evolving field of LLMs and AI. We ran the search string in the selected databases, resulting in 14 papers: seven in ArXiv, four in SCOPUS, two in ACM Digital Library, and one in IEEE Xplore. Finally, 12 papers were selected as the start-set for the snowballing when we read through the paper title, keywords, and abstracts. Table 1 presents these papers, ordered by the number of citations.

3.3. Snowballing forward and backward

The procedure to acquire papers for the literature review entailed both forward and backward snowballing of the starting set. Backward snowballing is achieved by looking at the references of each paper in the starting set to identify papers that can be likely candidates for the review. Forward snowballing is achieved by looking at papers that cited the reviewed paper to find more likely candidates for the review. For this work, the scraping of potential candidates was done exhaustively, using automated scripts (see Github repository) to include papers with a valid Digital Object Identifier (DOI) and manually for papers without a DOI. This scraping resulted in an initial set of 1583 papers.

3.3.1. Inclusion and exclusion criteria

Our study aims to identify metrics that are actionable for practitioners evaluating the quality of GenAI systems. Thus, we focused on empirical studies, such as GenAI applications in Engineering, FinTech, and Healthcare, which actively integrate GenAI models requiring evaluation. Studies evaluating image, video, or multimedia-based GenAI outputs were excluded, as we target evaluation metrics. With more empirical data, the metrics have a higher chance of being verified practically, and challenges or barriers might be more domain-specific, which lab studies may overlook.

We manually applied inclusion and exclusion criteria, shown in Table 2, to the title, abstract, and the result of each paper. After applying the inclusion and exclusion criteria, 69 papers were identified.

3.3.2. Rigor and industry relevance evaluation

To ensure the practical applicability of the identified metrics in industrial settings, we employed Ivarsson and Gorschek's **Rigor and Industrial Relevance** framework [45] to evaluate the selected papers. Motivations for choosing the framework:

Table 2
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Empirical studies, case studies, workshops	Theoretical papers without empirical data
Engineering, FinTech, Healthcare	Image, video files
Metrics evaluate outputs produced by GenAI systems	Studies do not include metrics
Qualitative or quantitative results	No conclusive results
Studies from 2014 to 2024	Studies before 2014
English language	Non-English language

Table 3
Scoring rubric for evaluating rigor [45].

Aspect	Strong description (1)	Medium description (0.5)	Weak description (0)
Context described	The context is fully detailed, enabling comparison with other contexts.	The context is mentioned, but lack detail for full understanding.	No context information is provided.
Study design described	The study design is explained in detail for understanding, including variables, controls, and sampling.	The study design is mentioned briefly with minimal details.	There is no description of the study design.
Validity discussed	The validity is thoroughly explained, including detailed threats and measures to address them.	Validity is mentioned but not explained in detail.	No mention of validity or its threats is provided.

Table 4
Scoring rubric [45] for evaluating industrial relevance.

Aspect	Contribute to relevance (score 1)	Do not contribute to relevance (score 0)
Subjects	The subjects: - Represent the intended uses of metrics - Industrial professionals	The subjects: - Students - Researchers - Not mentioned
Context	Industrial settings: metrics for the efficiency and effectiveness of GenAI systems. It covers: - Methods/approaches for the system qualities assessment. - General risks, challenges, or limitations.	Laboratory settings: - Not a real usage situation - Only model benchmarks - Other
Scale	Realistic industrial scale size	Toy examples
Research method	Research methods (RM) include: - Case study - Field study, orangeindustry survey/questionnaire - Action research - Lesson learned	Excluded RM: - Conceptual analysis - Conceptual mathematical - Laboratory experiment - N/A

- *Focus*: The framework supports the identification of concepts (metrics in the context of this study) that are both methodologically reliable and practically valuable.
- *Emphasis on industrial relevance*: It helps address the potential disconnect between reported metrics and their real-world applicability.
- *Quantitative scoring*: It offers a quantitative scoring rubric for rigor and relevance (see Table 4), allowing for more objective comparisons between studies and concepts (i.e. metrics).
- *Granular evaluation*: It provides a detailed review of industrial relevance through criteria including subjects, context, scale, and research method [45].

Procedure how Ivarsson and Gorschek’s framework was applied in this work:

1. The authors developed a rigor scoring rubric (see Table 3) based on Ivarsson and Gorschek’s framework.
2. The authors ran Ivarsson and Gorschek’s scoring rubric (see Table 4), focusing on practical applicability in industrial contexts.
3. All authors refined the scoring rubrics.
4. The authors randomly selected 5% of selected papers (3 out of 69).
5. The first, second, and third authors used the defined scoring rubrics and analyzed both rigor and relevance independently.
6. Rigor evaluation:

- (a) Read the full text of the selected paper, focusing on study contexts, design, and validity. Give the value of “1” for each of these three parts if they are detailed enough for full understanding or comparison; a value of “0.5” if they are partially described; and a value of “0” if there is no description.
- (b) Calculate the rigor scores using the formulas below.

$$Rigor = C + S + V$$

, where C refers to the context described, S is the study design described, and V stands for validity discussed. “Rigor” score ranges from zero to three, where a higher score indicates a more rigorous study.

7. Industrial relevance evaluation:

- (a) Read the full text of the selected paper, focusing on study subjects, context, scale, and research method (see Table 4).
- (b) Calculate Industrial Relevance scores using the formulas below.

$$Relevance = Subjects + Context + Scale + Researchmethod$$

For each item in the formula, assign “1” if the paper contains sufficient information, otherwise, assign “0”. The “Relevance” score ranges from zero to four, where a higher score indicates a better industrial-relevant study.

Table 5

A result of applying rigor and industrial relevance evaluation against the selected sample from the participated researchers.

ID	Paper title	Metric name	Include? (YES/NO)
Px	A discriminative metric for generation tasks with intrinsically diverse targets[S13]	DeltaBLEU	NO
Pm	Evaluating Large Language Models Trained on Code [S7]	PASS@k	YES
Pn	Evaluating the Factual Consistency of Abstractive Text Summarization[S39]	FactScore	YES

Table 6

An example of the table storing the result of rigor and industrial relevance analysis.

ID	Paper	Metric name	Rigor: context	Rigor: study design	Rigor: validity	Relevance: subjects	Relevance: Context	Relevance: Scale	Relevance: RM
Pm	[S7]	PASS@k	1	1	1	0	0	0	0
Pn	[S39]	FactScore	1	1	0	1	1	0	1

8. The authors recorded the scoring results for the evaluation set (3 out of 69), as shown in Table 5, from the participating authors.
9. Inter-Rater reliability [47] analysis. We used the Fleiss' Kappa coefficient [48] by following the guidelines from Landis and Koch [49] to assess how strong the level of agreement was among the authors. The result was $(\kappa) = .57, p < 0.05$, which represents "good" strength agreement among our authors' opinions.
10. When the researchers reached an agreement, the first author conducted the rigor and relevance analysis on all papers identified through the snowballing search.
11. The authors stored rigor and relevance analysis results in a spreadsheet file. An example is shown in Table 6.
12. The first author selected those papers with rigor scored "3" and industrial relevance scored "4", i.e. only papers that scored a "1" in all aspects of the model.
13. The authors reported the final result. Consequently, 43 papers were included and presented in Table A.15. These included papers are now cited as [S1]–[S43] to differentiate them from traditional references.

3.4. Data collection and analysis

Data collection involved two parallel processes: (I) Metrics (e.g., names, definitions, sample datasets, and implementation codes) were extracted using predefined codes (Level 1–3 in Fig. 2); (II) Risks and challenges were identified through inductive coding of qualitative data.

At each level, codes were developed by identifying recurring themes, concepts, and findings within the chosen papers.

- Level 1: GenAI system quality evaluation, risks or challenges of the evaluation. Four top-level codes were identified.
- Level 2: The Metric definitions, datasets, implementations, drawbacks or limitations, quality characteristics, known issues, concerns, and limitations. In total, 25 codes were used.
- Level 3: Metrics, top-level quality characteristics, evaluation method types. 121 codes were identified and analyzed at this level.
- Level 4: Sub-attributes of quality characteristics. 31 codes were used at this level.
- Supported papers: number of selected papers support the extracted codes.

An overview of data synthesis is presented in Fig. 3. We followed the below steps for the thematic analysis.

1. Creation of initial codes: Initially, we generated a starting set of codes, including metrics and quality characteristics. This initial set was informed by our research questions and the recurring themes observed in the selected papers.
2. Familiarization with extracted codes: We began the analysis by thoroughly reading and examining all extracted codes. At this stage, the codes reflected various aspects like metric definitions, features, methods, quality characteristics, and risks.

3. Following familiarization, we reviewed and refined the initial codes. This iterative process allowed for continuous refinement and accuracy in coding.
4. Connect and combine codes: After refining the codes, we began connecting and combining them to identify potential themes. By grouping related codes, we formed broader themes that encapsulated the key findings from the papers.
5. Finally, we summarized the contributions of the study based on the identified themes of collected metrics, methods, and risks. These themes present evidence that the study's insights and conclusions are derived from, ensuring that the findings were well-grounded in the data.

Consequently, we suggested an expanded framework inspired by Lavesson et al.'s original five-step model. This framework extends the model's steps and combines our findings with metrics' datasets, implementation, and practical evaluation processes. Details about the framework are presented in Section 5.

3.4.1. Steps for mapping metrics to quality characteristics

We present the steps for mapping metrics to ISO/IEC 25023 [22] quality characteristics. This mapping is essential for two reasons: (I) it allows us to refine the existing predefined measures in ISO/IEC 25023 if current metrics offer more precise or relevant assessments, especially for GenAI systems; (II) it enables us to complement the existing measures in cases where additional metrics are needed to address the evaluation of GenAI system outputs.

The general process of our mapping is presented in Table 7. To better explain the application of this mapping process, we selected the "METEOR [41]" metric, which is frequently used in evaluating GenAI system outputs, to illustrate the steps in detail.

Step 1: our mapping process begins with understanding a metric definition, and it is measured through a full-text reading of the selected papers. According to Banerjee et al.'s study [41], METEOR's main feature is to evaluate AI-generated texts' similarity while considering synonyms by comparing the generated output to human-written reference text.

Step 2: Examine the metric's measures and formula. In the example of METEOR, it measures $Precision = \frac{\text{Relevant retrieved instances}}{\text{All retrieved instances}}$ and $Recall = \frac{\text{Relevant retrieved instances}}{\text{All relevant instances}}$ of AI-generated texts with an additional penalty for incorrect word order. METEOR formula:

$$METEOR = \left(\frac{10 \times Precision \times Recall}{Recall + 9 \times Precision} \right) * \left(1 - 0.5 \times \left(\frac{\#chunks}{\#unigrams_matched} \right)^3 \right)$$

If a system generates the sentence "The cat sat on the mat", and the reference is "A cat sat on the rug". METEOR compares the words and accounts for synonyms (mat and rug) and stem forms (sat remains sat), resulting in a higher similarity score due to the semantic closeness.

Step 3: Compare metric functionality with ISO quality subcharacteristics' definitions. The METEOR metric's core function is to assess the semantic similarity of texts produced by GenAI systems, which indicates how suitable the system outputs are. The subcharacteristic

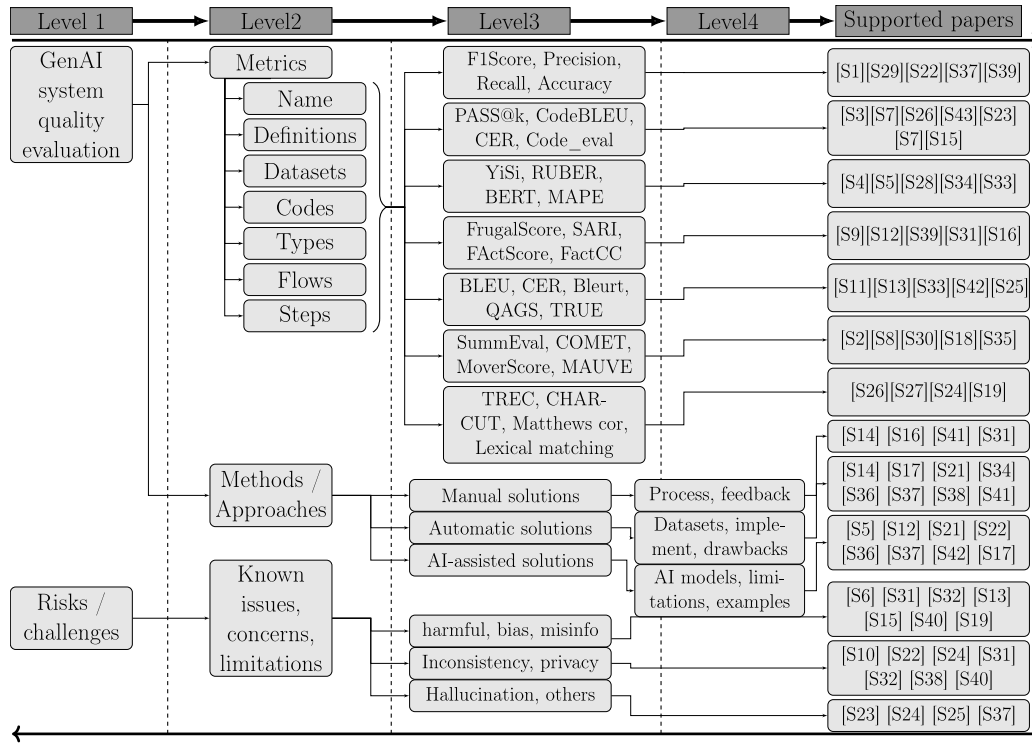


Fig. 2. Coding process for extracting data. Metrics were extracted via structured coding (Levels 2–3 at the top); Thematic analysis was applied inductively to identify risks/challenges (Levels 1–3 at the bottom).

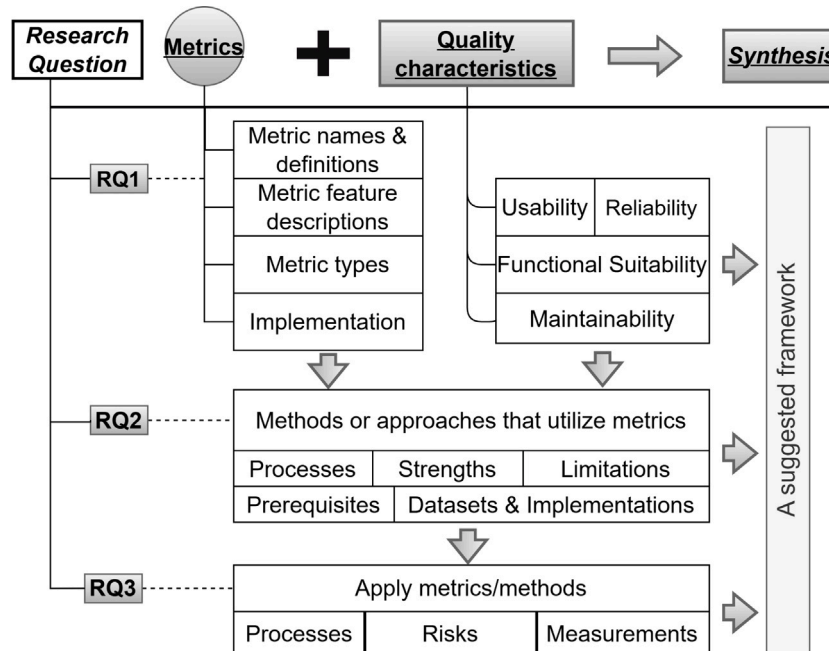


Fig. 3. An overview of data synthesis based on collected codes.

“Appropriateness recognizability” in ISO/IEC 25023 stands out as candidate mapping, according to its original definition: *Users have to be able to select a system/software product that is suitable for their intended use. The quality measures for appropriateness recognizability are used to assess the degree to which users can recognize whether a product or system is appropriate for their needs [22].*

Our interpretation of this subcharacteristic in the context of GenAI is: Appropriateness recognizability refers to how well the generated

outputs align with their intended use, indicating whether engineers can determine if the output is suitable for their specific needs (e.g., document generation).

This interpretation explains why we mapped the METEOR metric to “Appropriateness recognizability” as an example. It is important to note that while the original definition focuses on systems or components, our interpretation centers on GenAI system outputs, considering the inherent black-box nature of GenAI models.

Table 7
Steps for mapping the identified metrics to quality characteristics and subcharacteristics predefined by ISO/IEC 25023.

Steps	Description
Step 1	Read a metric feature description.
Step 2	Check the metric's measures and learn its formula, if available.
Step 3	Compare the metric's functionality with quality subcharacteristics definitions, identifying relevant subcharacteristics by evaluating how the metric assesses AI-generated outputs. For example, how relevant the outputs are.
Step 4	Map the metric to quality characteristics by examining how the link supports overall system qualities that have been predefined by ISO 25023.
Step 5	Exclude any mappings that are not supported by the reported use cases or evidence from the selected paper.
Step 6	Finalize the mapping of the metric to predefined quality characteristics through linked subcharacteristics.

Step 4: Map the metric to quality characteristics. In ISO/IEC 25023, “Usability” is the top level quality characteristic for “Appropriateness recognizability”. METEOR is mapped to Usability since the metric result indicates to which degree the system-generated text is understandable.

Step 5: Exclude any unsupported mappings. In Step 3, METEOR could map to the “Maturity” subcharacteristic, according to its original definition: *Maturity measures are used to assess the degree to which a system, product, or component meets the needs for reliability under normal operation* [22]. In the context of GenAI, we interpreted Maturity as reflecting the system’s ability to consistently produce expected outputs under normal conditions, which aligns with the original definition. However, our review of the selected papers revealed insufficient direct evidence to support this link. Therefore, this mapping was excluded.

Step 6: Finalize the mapping of the metric to predefined quality characteristics through the associated subcharacteristics. In this example, METEOR is ultimately mapped to Usability through the subcharacteristic “Appropriateness recognizability”.

Notably, although we have provided one example, the mapping process involved analyzing various other characteristics during the data analysis phase. By detailing each step, we aim to show transparency in the mapping process and elaborate the logic behind connecting a metric to specific quality characteristics. To minimize bias and potential misinterpretations, three authors actively participated in the development and analysis of this mapping through regular discussions and reviews.

3.5. Steps for identifying evaluation methods

The evaluation methods for answering RQ2 were derived by following below steps.

- Step 1:** Extracting evaluation processes. While collecting metrics from the literature, we extracted description of evaluation processes from the selected papers.
- Step 2:** Thematic coding. Level 3 codes, such as “human evaluation”, “automated evaluation”, and “AI-assisted evaluation” were created from phrases.
 - Human evaluation: Final assessment is human-driven, even if automated tools are used in data preparation.
 - Automated evaluation: Algorithm-driven scoring without human involvement.
 - AI-Assisted evaluation: Using AI models to augment human judgment or automate scoring.
- Step 3:** Linking codes to metrics and paper references. Metrics and papers were grouped based on their primary evaluation methods. For example, papers using METEOR without domain-experts involvement were coded as automated evaluation. Studies employing automated metrics using AI models were coded as AI-assisted evaluation.

- Step 4:** Validation. Three authors of this study cross-validated the coding. Discrepancies (e.g., ambiguous cases) were resolved through multiple group discussions.

3.6. Synthesizing processes for risky output evaluation

The processes for RQ3 about risky output evaluation were synthesized through:

- Collecting code:** Coding recurring patterns (e.g., risk mitigation steps, dataset validation) from the selected papers. The themes include Level 3 codes (see Fig. 2 including bias or harmful content).
- Common process:** Grouping similar processes, for example: “steps validate system outputs against reference datasets” → Dataset preparation; “tracking system quality thresholds” → Monitoring.
- Synthesis of practices:** Analyzing papers that implicitly described steps without formalizing workflows. For example, FActScore [26] was used for fact-checking but did not outline a process.
- Reviews:** Reviewing coded themes iteratively into a generalized process by three authors, ensuring alignment with the best practices of system outputs’ evaluation.

4. Results

This section presents study results to answer research questions.

4.1. RQ1: What metrics can be used to evaluate the quality characteristics defined by ISO/IEC 25023 for generative AI system outputs?

We have identified 28 distinct metrics, as shown in Table 8. We observed that metrics like METEOR are commonly used to evaluate text similarity — how closely AI-generated text aligns with natural language and meaning. Additionally, Lexical matching, CER, and BERTScore assess low-level relevance, such as word, character, and token similarity. Metrics like FActScore and TRUE focus on evaluating the factual accuracy of generated text. These metrics reflect growing concerns about both linguistic and factual accuracy in AI-generated outputs.

In the field of code generation, metrics such as Code_eval and CodeBLEU are infrequently used in the literature, possibly indicating that the development of robust evaluation for this domain is still ongoing.

Additionally, metrics such as COMET and YiSi, which utilize contextualized embeddings, aim to evaluate GenAI systems integrated with enterprise proprietary data stored in vectorized databases. These embedding-based metrics are particularly valuable for use cases where pre-trained LLMs may not meet the specific requirements imposed by private data.

Table 8

Identified metrics, feature description, supporting papers, and target usage domains. (X) - Number of supporting papers for a metric. (Embeddings) - Metric can process embeddings as contextualized data.

Metric name	Feature description	Supported paper(s)	Usage domain(s)
BLEU	Measures n-gram overlap between texts	[S3] [S33] [S42] [S16] [S13] [S5] [S4] [S6] [S7] [S10] [S9] [S25] (12)	Text similarity Text matching
METEOR	Evaluates precision and recall in text matching	[S35] [S34] [S4] [S25] [S14] [S10] [S8] [S30] (8)	Text similarity Synonymy matching
F1Score	Special mean of precision and recall	[S1] [S4] [S22] [S29] [S33](5)	Text segmentation Classification
BERTScore	Compares embeddings for semantic similarity	[S20] [S4] [S21] [S9] [S2] (5)	Token similarity
Precision	Ratio of correct positive predictions	[S29] [S29] [S22] (3)	Data retrieval
Recall	Ratio of correctly identified true positives	[S29] [S22] [S29] (3)	
Bleurt	Evaluates text fluency and semantic similarity	[S11] [S21] (2)	Semantic similarity
Matthews correlation	Assesses classification accuracy for imbalanced data	[S24] [S42] (2)	Classification Linear correlation
Pearson correlation	Measures linear correlation between variables	[S14] [S42] (2)	
Code_eval	Tests code correctness	[S7] [S33] (2)	Code correctness
YiSi	Evaluates translation quality using multilingual embeddings	[S28] [S31] (2)	Semantic similarity (Embeddings)
Lexical matching	Matches exact words between texts	[S19] [S7] (2)	Word similarity
COMET	Evaluates translation quality with contextual embeddings	[S8](1)	Semantic similarity (Embeddings)
MAUVE	Assesses how close AI-generated text is to human writing	[S18](1)	Text analysis Human-like texts
RUBER	Measures the similarity between generated texts and references	[S4](1)	Text similarity (Embeddings)
MAPE	Measures accuracy of predictions in regression tasks	[S5](1)	Numerical accuracy Char accuracy
CER	Calculates character-level error rate	[S23] (1)	
TREC	Evaluates the quality of QA outputs with multiple datasets	[S26](1)	Text accuracy Text relevance
PASS@k	Checks how often code passes tests within k attempts	[S3](1)	Code correctness
FrugalScore	Evaluates semantic similarity	[S9](1)	Semantic similarity
SummEval	Assesses summary quality using multiple evaluation models	[S2](1)	Text summarization
TRUE	Measures QA accuracy using truthfulness and reasoning	[S10](1)	Factual consistency
MoverScore	Measures semantic similarity with word embeddings	[S30](1)	Semantic similarity (Embeddings)
FActScore	Evaluates factual consistency in generated text	[S39](1)	Factual consistency
QAGS	Checks for factual correctness in summaries	[S32](1)	Text summarization
CHARCUT	Measures character-level changes between texts	[S27](1)	Char matching
SARI	Evaluates quality of text simplifications	[S12](1)	Text accuracy
CodeBLEU	Assesses code generation with structure and syntax checks	[S43](1)	Code syntax matching

4.1.1. Mapping metrics to ISO quality characteristics

To gain a finer-grained view of how the identified metrics contribute to the overall quality of generative AI systems, we mapped them to ISO/IEC 25023 quality characteristics, as visualized in Fig. 4.

As shown in Fig. 4, the identified metrics correspond to four main characteristics: usability, reliability, functional suitability, and maintainability. These mappings are built by using the process described in

Section 3.4.1. Metrics (e.g., METEOR and F1Score) are fundamentally based on Precision and Recall, and BLEURT is derived from BLEU. To maintain focus and avoid redundancy, only top-level metrics are included in our mapping.

Building upon Hadi et al.'s categorization [7] and the reported cases in the selected papers, we identified six key GenAI application domains: document generation, data analysis and insights, marketing

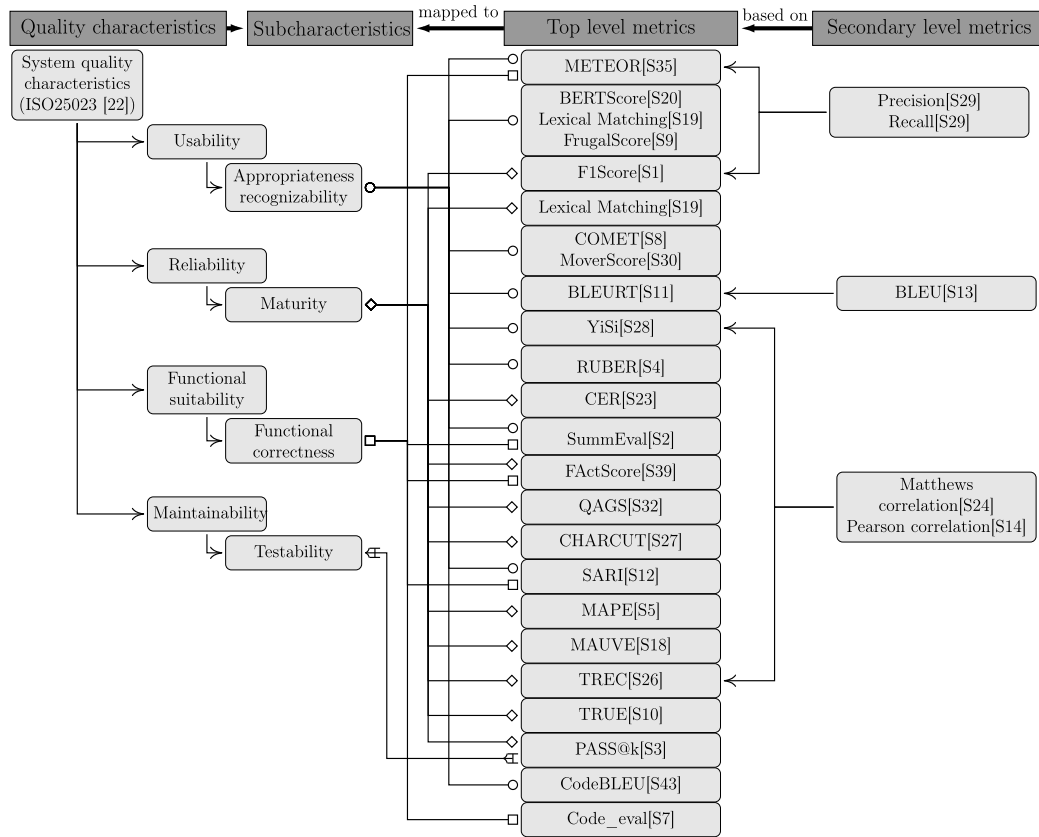


Fig. 4. Mapping the identified metrics to ISO/IEC 25023 quality characteristics.

Table 9

Metrics mapped to quality characteristics and GenAI application domains.

Quality characteristics	Sub-characteristics	GenAI application domains						SUM
		Document generation	Data analysis and insights generation	Marketing and content creation	Customer service ChatBots	Multilingual business communication	Code generation	
Usability	Appropriateness recognizability	METEOR BLEURT BERTScore MoverScore FrugalScore YiSi COMET SARI	SummEval METEOR BLEURT BERTScore MoverScore YiSi	METEOR BLEURT BERTScore MoverScore	BLEURT RUBER	RUBER	CodeBLEU	11
Reliability	Maturity	FActScore QAGS CHARCUT Lexical matching	TRUE F1Score	QAGS MAPE	F1Score TREC	CER MAUVE F1Score	PASS@k	11
Functional suitability	Functional correctness	METEOR SummEval FActScore	SummEval	SARI	–	–	Code_eval	5
Maintainability	Testability	–	–	–	–	–	PASS@k	1
SUM (The number of metrics)		13	8	7	4	4	3	–

and content creation, customer service chatbots, multilingual business communication, and code generation. These domains represent areas where LLMs have demonstrated significant business impacts, particularly in enterprise contexts [50]. Table 9 presents how the metrics map to quality characteristics within these application domains.

Usability: The identified metrics like METEOR, BLEURT, BERTScore, MoverScore, FrugalScore, YiSi, COMET, SARI, and RUBER,

are mapped to “Appropriateness recognizability”, as they assess how closely the generated outputs align with intended use cases across domains (e.g., document generation or content creation). For code generation tasks, CodeBLEU evaluates the appropriateness of AI-generated code.

Reliability: Metrics, such as FActScore, QAGS, CHARCUT, Lexical matching, TRUE, and MAPE, are mapped to “Maturity”, reflecting the

GenAI system’s ability for content creation to produce accurate outputs under normal conditions. Metrics like F1Score, and TREC provide precision measures, critical for ensuring factual consistency in customer service chatbot applications. Metrics, such as CER and MAUVE, offer measurements to ensure accurate human-like texts for multilingual business communication, and PASS@k measures the GenAI system’s ability to consistently produce accurate code solutions.

Functional suitability: Metrics, such as SummEval, SARI, FActScore, METEOR, and Code_eval, map to “Functional correctness”, as they assess how accurately the GenAI system outputs meet task-specific requirements, such as summarization or text simplification, indicating the system’s ability to deliver correct results for the intended use.

Maintainability: PASS@k is mapped to “Testability”, reflecting the ease with which the system’s output can be verified. This is particularly useful for code generation, where the metric indicates how many trials are needed to produce accurate code, thus providing an indicator of how effectively the GenAI system can meet predefined criteria.

Key observations:

- Our mapping reveals that Usability and Reliability are the most frequently addressed quality characteristics in the identified GenAI application domains. This emphasizes the critical aspects for the GenAI system to generate outputs that are both appropriate and consistent.
- Functional suitability is primarily addressed in domains where the correctness of the output is critical, such as document generation.
- Maintainability, while less frequently mapped, plays a vital role in ensuring that generated code can be efficiently tested and validated, highlighting the importance of evaluating long-term system qualities for code generation tasks.
- Certain metrics address multiple ISO characteristics depending on their application context. For example, PASS@k evaluates both code maturity (Reliability) and testability (Maintainability). While this overlap reflects the metric’s versatility, practitioners should interpret results based on their specific quality priorities.

While our study mapped metrics to four ISO/IEC 25023 quality characteristics, gaps remain in addressing the other characteristics, for example, Performance Efficiency and Security particularly. These characteristics are critical for industrial adoption but lack dedicated metrics in current literature.

In conclusion, our findings — from a list of available metrics, through their mapping against ISO quality characteristics, to their alignment with the reported application domains — offer a matrix view on the use of metrics to assess generative AI system qualities. The key takeaway is the importance of selecting metrics that are closely aligned with specific system qualities and relevant to each GenAI application domain.

4.2. RQ2: What methods or approaches exist that could utilize the identified metrics to assess the quality of outputs produced by generative AI systems?

Analysis of the identified metrics shows that they can be used in different methods. In this section, we have presented a synthesis of these methods and connected them to the different metrics—Results from RQ1. Organizations might apply these metrics directly without realizing they are inherently tied to a particular method. This flexibility means that while metrics can be used independently, understanding the methods behind them provides insights into the processes, strengths, and limitations of the chosen evaluation approaches.

By following the steps in Section 3.5, three evaluation methods, such as human, automated, and AI-assisted, were identified from the literature. For instance, three metrics in Table 10 described human evaluators validating system outputs, while 21 metrics in Table 11 utilized scripts for automated scoring. Five AI-assisted metrics in Table 12

Table 10
Metrics from RQ1 that can support human evaluation.

Metric from RQ1	Evaluation	Example
SARI	Outputs with citations	Measuring text similarity extracted from documents
PASS@k	Code generation review	Assessing human reviews for generated codes
SummEval	Review report	Evaluating reviews’ summary

were reported using model-based evaluators. Details of these evaluation methods are presented separately in Sections Section 4.2.1, 4.2.2, and 4.2.3.

4.2.1. Human evaluation method

Human evaluation involves the manual review and assessment of outputs produced by GenAI systems, typically conducted by domain experts or end-users. This method ensures that outputs are not only technically accurate but also contextually appropriate for industry-specific applications.

Evaluating outputs with domain experts offers a deeper understanding of industry contexts and facilitates detailed error analysis. For example, while GenAI systems may produce code that compiles without errors, human evaluators can assess whether the code adheres to best practices and company standards and integrates with existing codebases. This evaluation ensures that functional outputs also align with design patterns critical for team collaboration.

The suggested evaluation process, illustrated in Fig. 5, includes the following steps:

1. Industrial contexts: The process begins with enterprise data inputs, which can be documents or requirements with context information.
2. Prompts: The data inputs are then fed into an GenAI application with prompts under test.
3. Outputs: Responses or results are generated by GenAI applications based on the input prompts and user queries.
4. Feedback: The outputs are then passed to human experts with domain knowledge or competencies. These experts review the outputs and provide feedback. Metrics could be scores about outputs’ similarity or percentage of expert judgment.
5. Results: Human feedback contributes to the generation of final results, which may include quantitative data towards the feedback, such as the percentage of agreement among feedback providers.

Table 10 shows that metrics can play a supportive role by providing a quantitative indicator that human evaluators can use to guide their assessments. For instance, metrics such as SARI can help reduce the cognitive load on human evaluators by filtering outputs that meet a threshold score of 0.8 in semantic similarity and providing cited source data. This allows evaluators to focus on a smaller set of high-quality results. These metrics are classified under human evaluation because the final assessment relies on domain-experts’ judgment.

While human evaluation remains essential, it is not immune to errors, as evaluators could miss or misinterpret risky outputs. By automating the filtering process with metrics, we can streamline the evaluation and provide an objective layer of validation, thereby reducing the likelihood of human bias. This approach ensures that human judgments are supported by data, further enhancing the final assessment.

For practitioners, human evaluation requires evaluators to possess domain-specific knowledge. **For those evaluators with limited validation capabilities, unverified outputs could introduce risks without proper validation.** Additionally, scaling human evaluation for large datasets is challenging, highlighting the necessity for automated methods, which we explore in the next section.

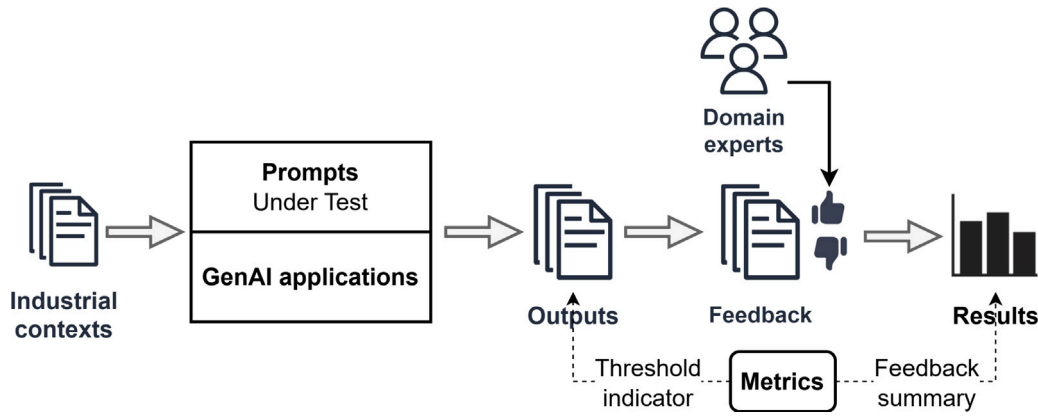


Fig. 5. A suggested human evaluation process using metrics.

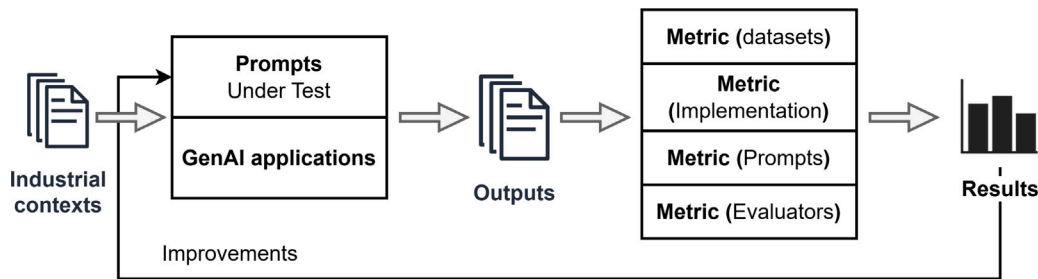


Fig. 6. An identified evaluation process in automatic manners.

4.2.2. Automated evaluation method

Automated evaluation involves using algorithms, scripts, and metrics to assess the quality of AI-generated outputs and enables scalable and efficient evaluation across large datasets.

Automated evaluation methods provide consistent and repeatable assessments with predefined standards [36]. These methods are typically used for identifying patterns and trends in AI-generated outputs, making them suitable for evaluating large volumes of data where human evaluation may be impractical.

The evaluation process, as illustrated in Fig. 6, involves several key components and steps:

1. Industrial contexts input: The process begins with the identification and input of relevant industrial contexts. These contexts define the requirements, constraints, and specific needs that the GenAI applications must meet.
2. GenAI applications and prompts under test: GenAI applications are then tested using various prompts that align with the industrial contexts. These prompts are designed to simulate real-world scenarios.
3. Generating outputs: The application processes the prompts and generates outputs. These outputs are the direct results of the LLM's response to the provided prompts within the context of the defined industrial needs.
4. Application of metrics: The outputs are then evaluated using metrics.

- (a) Metric (datasets): The outputs are compared against relevant datasets to validate their correctness, accuracy, and relevance. These datasets provide a reference against which the LLM applications' output is measured.
- (b) Metric (Implementation): This refers to the practical application of the metrics. For example, this might involve using Python scripts to automatically calculate metric

scores, ensuring that the evaluation process is both executable and reproducible.

- (c) Metric (Prompts): Prompts used to assess GenAI applications.
- (d) Metric (Evaluators): Evaluators (e.g., algorithms/scripts) involve a qualitative and quantitative assessment to ensure that the outputs meet the defined quality standards.

5. Results analysis: The results from these evaluations are then compiled and analyzed to determine how well the GenAI application has performed in the context of the industrial tasks. While metrics provide a quantitative assessment, an expert or human evaluator is important in interpreting and monitoring the metric data.

6. Feedback Loop for Improvements: The insights gained from the results analysis are fed back into the process, informing improvements to the GenAI applications, the design of prompts, and the overall evaluation process.

Table 11 provides metric implementations, along with sample data and GitHub repository links. When implementing a metric, practitioners may need to interact with the code, possibly adapt it for their particular scenarios, and verify that their datasets conform to the metric's prerequisites.

While automated evaluation provides scalability and consistency, there are inherent limitations:

- Contextual Limitations: Metrics like CER are highly context-dependent and may not fully capture the content they are evaluating. For example, CER focuses on character-level accuracy, which is useful for technical content but might miss broader errors in context or logic.
- Dependence on Reference Data: Many metrics, including CER, rely heavily on high-quality reference data. The results of these metrics depend on the availability of accurate and comprehensive

Table 11**Metrics** from RQ1 with prerequisites, datasets, implementations, and drawbacks/limitations for automated evaluation.

Metric from RQ1	Prerequisites	Datasets & Implementation	Drawbacks/Limitations
BLEU	– Reference translations. – Candidate translations.	– WMT14. – NLTK BLEU.	– Limit to n-gram matches. – Overlook semantic meaning. – Sensitive to minor variations
METEOR	– Reference translations. – Candidate translations.	– WMT19. – Meteor Github	– Depend on high-quality references. – Penalize creative paraphrasing.
F1Score, Precision, Recall	– Ground truth labels. – Predicted labels	– Huggingface. – Scikit-learn Github.	– Misleading for imbalanced datasets. – Does not account for true negatives.
Matthews correlation	– Ground truth labels. – Predicted labels.	– Kaggle data. – Github.	– Sensitive to class imbalance. – Require confusion matrix.
Pearson correlation	– Paired numerical data points.	– Kaggle data. – SciPy Github.	– Assume linear relationships. – Sensitive to outliers. – Does not imply causation.
Code_eval	– Reference code. – Generated code. – Test cases.	– HumanEval. – Codex Github.	– Focus on test case pass rate. – Require extensive test cases. – Limit to functional correctness.
YiSi	– Source texts. – Reference translations. – Candidate translations.	– WMT16. – YiSi Github	– Requires multilingual embeddings. – Sensitive to translation quality. – Complex to implement.
Lexical Matching	– Reference texts. – Candidate texts.	– KaggleNLP. – NLTK Github.	– Rely on exact matches. – Overly simplistic for complex texts.
MAPE	– Predicted values. – Actual values.	– Kaggle data. – Sklearn Github.	– Sensitive to small errors. – Biased by outliers. – Less effective for zero or near-zero values.
CER	– Reference texts. – Hypothesis texts.	– Openslr Huggingface. – CER Github.	– Focus on character-level errors. – Overly penalize minor typos. – Not suitable for longer texts.
TREC	– Data samples.	– Trectools data. – TREC Github.	– Focused on information retrieval.
RUBER	– Data samples.	– datasets. – RUBER Github.	– Focused on dialogue systems.
PASS@k	– Problem descriptions. – Generated code samples. – Unit tests.	– HumanEval. – PASS@k Github.	– Dependent on test case quality. – Does not assess code efficiency. – Limited to functional correctness.
BERTScore	– Reference texts. – Hypothesis texts.	– Huggingface data. – BERTScore Github.	– Maximum sentence length is 512.
FrugalScore	– Small set of human-annotated translations.	– Kaggle NLP – FrugalScore Github.	– Potential bias from small dataset. – Scaling issues with large datasets. – Limited semantic range capture.
SummEval	– Reference summaries. – Candidate summaries. – Multiple pre-trained evaluation models.	– CNN/Daily mail. – SummEval Github.	– Dependent on multiple models. – Complexity in aggregation. – High computational demand.
FActScore	– Source documents. – Generated summaries.	– Wiki data. – FActScore Github.	– Focused on factual consistency only. – Sensitive to source quality. – Require fact extraction.
CHARCUT	– Reference texts. – Hypothesis texts.	– WMT17. – CHARCUT Github.	– Focused on character-level edits. – Not suitable for complex texts.
SARI	– Original texts. – Simplified reference texts.	– WikiLarge. – SARI Github.	– Require high-quality references. – Penalize creative simplifications.
QAGS	– Generate questions. – Answer questions. – Compare answers.	– NewsQA. – QAGS Github.	– Prepare QA data.
CodeBLEU	– Reference code. – Generated code.	– CodeXGLUE. – CodeBLEU Github.	– Complex scoring system. – Focused on code structure. – Require high-quality reference code.

reference datasets. In cases where such data is limited or biased, the metrics might provide inaccurate or less reliable evaluations.

– Inability to Handle Ambiguity: Metrics like SARI are not well-suited to handling ambiguous prompts where multiple correct

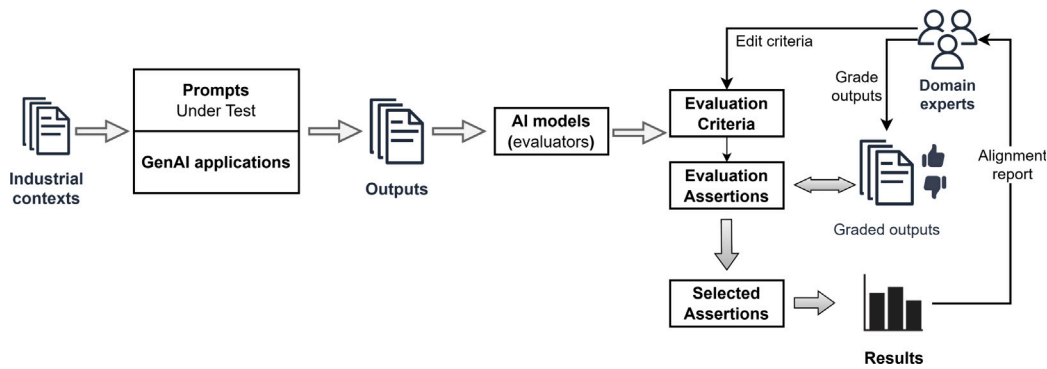


Fig. 7. A synthesized AI-assisted evaluation process based on literature. Domain experts iterate through the process to edit evaluation criteria and system output grading.

answers may exist. These metrics are often rigid in their assessment criteria, which can lead to penalizing outputs that are correct but expressed differently than the reference data.

Automated evaluation offers a scalable approach for evaluating GenAI system outputs, but practitioners must be aware of its limitations.

4.2.3. AI-assisted evaluation method

AI-assisted evaluation involves using AI models to automate the initial assessment of system outputs, process results, and suggest refinements, thereby enhancing and adapting the overall evaluation process.

As illustrated in Fig. 7, the AI-assisted evaluation process offers an adaptive assessment that refines evaluation criteria over time to maintain the quality of GenAI system outputs. The process involves the following key steps:

1. Industrial contexts: Input industrial contextualized information or documents to the GenAI system.
2. Prompts Under Test and GenAI applications: Industrial contexts are processed through the prompts in GenAI applications.
3. Generating Outputs: The GenAI applications generate outputs based on the input data and prompts.
4. AI Evaluators: AI model(s) trained for evaluation, assess the outputs.
5. Evaluation Criteria: The evaluator LLMs use predefined criteria to judge the outputs.
6. Evaluation Assertions: Based on the criteria, the evaluator models generate specific assertions about the quality of the outputs.
7. Graded Outputs: The original outputs are graded based on these assertions.
8. Domain Expert Involvement: Domain experts review the graded outputs and can refine the evaluation criteria and grading process.
9. Results and Alignment Report: The process produces evaluation results and an alignment report, which can be used to improve the system.

To implement AI-assisted evaluation, selecting appropriate metrics and AI models is important. Table 12 presents a detailed overview of metrics, automation steps, required AI models, available datasets, and implementation examples.

Table 12 highlights that metrics can utilize pre-trained AI models like GPT-2, showcasing the versatility of LLMs in evaluation tasks [51].

AI-assisted evaluation methods offer several advantages over automated metrics:

- Contextualized embeddings: AI-assisted metrics use pre-trained language models to create context-aware representations of text. This way can capture the subtleties of meaning and intent, rather

than relying solely on exact word matches. However, this approach requires access to context-dependent data, which can be challenging.

- Human-aligned learning: Many of these metrics are fine-tuned on datasets of human ratings, allowing them to better align with human perceptions of quality.
- Semantic comprehension: These metrics excel at capturing semantic similarities, going beyond surface-level text features. By understanding the underlying meaning of a text, AI-assisted metrics provide more accurate and meaningful assessments of generated outputs.

Despite their advantages, AI-assisted evaluation methods face certain limitations:

- Dependence on metric dataset quality: These metrics rely heavily on high-quality, contextually relevant datasets for evaluation. Acquiring such datasets can be challenging in financial analysis, where data is often sensitive and difficult to obtain.
- Complexity in implementation: AI-assisted metric implementation may require extra technical expertise compared to automated metrics. The need to integrate and deploy AI evaluation models, such as COMET, can be a barrier for teams.
- Interpretability challenges: The outputs from AI-assisted evaluation may be less transparent, potentially making it difficult for evaluators to understand the reasoning behind a given score in the evaluation process.
- Computational intensity: The advanced processing capabilities of AI-assisted metrics often come at the cost of higher computational requirements. Running these metrics can be resource-intensive, making them less suitable for scenarios where computational resources are limited.

4.2.4. Summary of human, automated, and AI-assisted evaluation

Human evaluation, despite being resource-intensive and subject to individual biases, provides detailed and context-aware assessments for evaluating GenAI system outputs. When integrated with automated and AI-assisted evaluation techniques, it enhances the robustness and depth of the quality assurance process.

For the simplification of analytical clarity, we classify evaluation methods into three categories — human, automated, and AI-assisted. In real-world cases, the degree of automation in evaluation can vary. Feldt et al. [52] highlight their view in a taxonomy on the ways AI is applied in software engineering, distinguishing between levels such as automation support, decision augmentation, and full automation.

For example, in AI-assisted evaluation, domain experts often iteratively refine prompts while relying on AI-generated assessments as input. This interaction aligns more closely with the “collaboration” levels described in Feldt et al.’s taxonomy, rather than full automation. Similarly, some automated methods embed indirect human assumptions

Table 12

Metrics from RQ1 with required AI models, datasets, implementations, and examples that can be used for AI-assisted evaluation.

Metric from RQ1	AI assist	Datasets & Implementation	Example	Drawbacks / Limitations
Bleurt	BLEURT	– WMT21. – Bleurt Github	<pre>from bleurt import score references = ["The cat sits on the mat."] candidates = ["There is a cat on the mat."] scores = score.score(references=references, candidates=candidates) print(f"BLEURT Score: {scores[0]:.3f}")</pre>	– Requires extensive fine-tuning. – Sensitive to domain shifts. – High computation cost.
MAUVE	GPT2	– GPT-2 data. – MAUVE Github.	<pre>from examples import load_gpt2_dataset import mauve p_text = load_gpt2_dataset('data/amazon.valid.jsonl', num_examples=100) # human q_text = load_gpt2_dataset('data/amazon-xl-1542M.valid.jsonl', num_examples=100) # machine out = mauve.compute_mauve(p_text=p_text, q_text=q_text, device_id=0, max_text_length=256, verbose=False) print(out.mauve) # prints 0.9917</pre>	– Relies on large datasets. – May not capture all nuances of language quality. – Requires high computation resources.
MoverScore	BERT Embeddings	– WMT19. – MoverScore Github.	<pre>from moverscore_v2 import sentence_score refs = ['The dog bit the man.', 'The dog had bit the man.'] sys = 'The dog bit the man.' moverscore = sentence_score(sys, refs) print(f"Mover Score: {moverscore[0]:.3f}")</pre>	– Dependent on high-quality word embeddings. – Limited by the quality of the reference text.
TRUE	T5	– Annotation sentences. – TRUE Github.	<pre>python true/src/meta_evaluation_scores.py --input_path=data/"(INPUT_FILE)" \ --metrics_scores_columns="{METRICS}" \ --output_path="{OUTPUT_PATH}"</pre>	– Focused on factual consistency. – Require specific data types.
COMET	COMET	– WMT22. – COMET Github	<pre>from comet import download_model, load_from_checkpoint model_path = download_model('wmt20-comet-da') model = load_from_checkpoint(model_path) data = [{"src": "The cat sits on the pad.", "mt": "There is a cat on the pad." "ref": "A cat is sitting on the pad."}] seg_scores, sys_score = model.predict(data, batch_size=8, gpus=1) print(f"COMET Score: {seg_scores[0]:.3f}")</pre>	– Requires complex model fine-tuning. – Sensitive to the quality of input encodings. – High resource demands for processing.

through dataset design or threshold calibration, blurring categorical boundaries.

While our findings help guide the practical usage of evaluation methods, we acknowledge that evaluation methods exist along a continuum of automation rather than as strictly evaluation types. As evaluation tools mature and human–AI collaboration increases, future work should consider extending the current classification of quality evaluation methods for GenAI systems.

4.3. RQ3: What processes can be used to apply metrics for evaluating the potential of risky outputs generated by generative AI systems?

GenAI systems can generate outputs that are misleading, biased, and inaccurate. Through this study, we identified six types of risky outputs, as shown in [Table 13](#): harmful content, bias and discrimination, misinformation, inconsistency, privacy violations, and copyright infringement.

Metrics can be used to measure a specific risk. For instance, the MAUVE metric can be utilized to assess harmful content; FActScore or TRUE metrics can be employed to check the misinformation. These measurements can flag potentially risky outputs before they cause harm in real-world applications.

Following our predefined steps (presented in [Section 3.6](#)), we synthesized a three-step process for risky output evaluation. As illustrated in [Fig. 8](#), the process contains three steps: Step 1 - prepare datasets, Step 2 - compute scores, and Step 3 - monitor scores.

Step 1: prepare datasets: This step involves gathering datasets that represent the GenAI system's intended application domain. Practitioners should incorporate sample datasets that reflect diverse contexts and can be extended to develop enterprise-specific datasets, ensuring evaluations are relevant to their business scenarios.

In Step 2: compute scores, Using the prepared datasets, outputs are generated from the GenAI models and evaluated against predefined metric prompts. This scoring quantifies key aspects such as accuracy, safety, and bias. Baseline scores are established from standalone AI

Table 13
Risky output dimensions, example metrics, and measurements.

Risk dimensions	Example metrics	Measurements
Harmful Content	MAUVE	Measures the degree of toxicity in outputs, helping identify potentially harmful content.
Bias and Discrimination	BERTScore, BLEURT, COMET	Analyzes outputs for demographic or ideological biases using semantic similarity and contextual understanding.
Misinformation	FactScore, QAGS, TRUE	Compares outputs against trusted sources to detect false information.
Inconsistency	F1Score, Precision, Recall	Measures consistency across multiple system outputs for the same input.
Privacy Violations	Lexical Matching, TRUE	Uses adversarial testing and named entity recognition to identify potential disclosure of sensitive information.
Copyright Infringement	BLEU, METEOR, BERTScore, MoverScore	Compares outputs to reference texts to detect potential plagiarism.

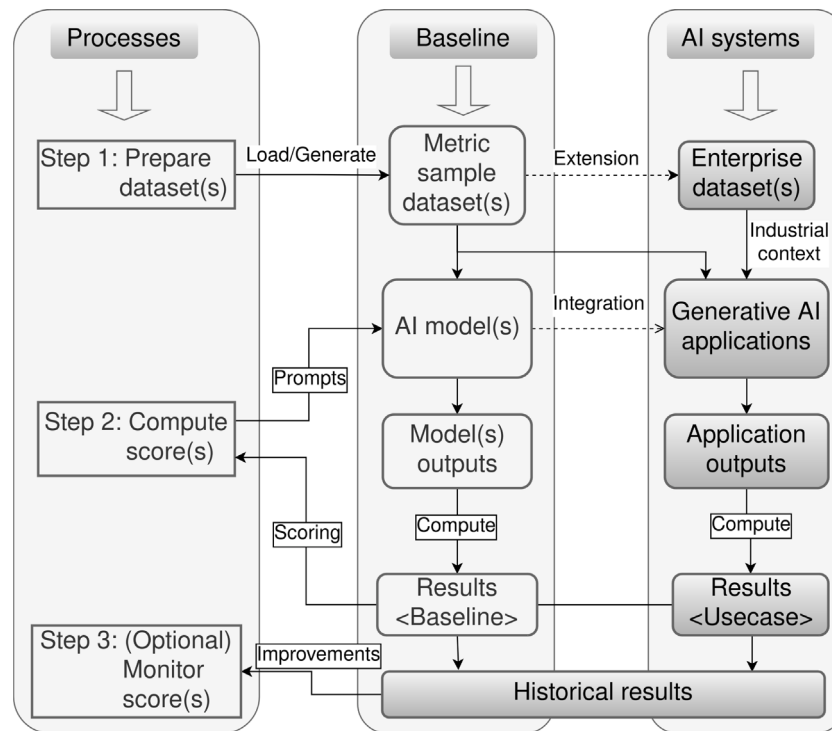


Fig. 8. A three-step process to apply metrics for evaluating the risky outputs produced by generative AI systems.

models, while scores from GenAI applications provide performance insights in specific use cases. In industrial contexts, proprietary datasets can be challenging to develop; in such cases, sample datasets may be utilized for evaluations.

Step 3: monitor scores (Optional): Monitoring involves continuously tracking metric scores over time to detect variations in the quality of GenAI outputs. By comparing results to the established baseline, organizations can identify trends and potential areas needing improvement. This step is particularly valuable for long-term quality assurance and allows for prompt detection and mitigation of emerging risks.

To illustrate the processes, an example using BERTScore [19] metric to assess gender biases in AI-generated outputs is provided. Steps are:

1. **Step 1: Prepare datasets.** Datasets should include prompts that represent a wide range of demographic groups and contexts. One can utilize an existing dataset or create a new one. For example, a [dataset](#) about profession, gender, and roles might include sentences like “The programmer completed the code”,

“The athlete secured the victory”, or “The doctor instructed the nurse”.

2. **Step 2: Compute Scores.** The prepared datasets, along with prompts, are inputted into the AI model (such as GPT), and the outputs produced by the model are collected. The BERTScore metric calculates scores by comparing the AI-generated outputs against a set of unbiased reference outputs. For example, if the prompt is “The doctor said to the nurse”, and the AI consistently generates “he” for the doctor and “she” for the nurse, it may suggest a gender bias.
3. **Step 3: Monitor Scores.** Track BERTScore over time to detect deviations indicating potential biases. If biases are identified, refine the model by adjusting datasets or prompts and recompute the scores to assess improvements.

While this step is optional, it is essential to periodically re-evaluate the GenAI system outputs displayed in trend visualizations with updated datasets and prompts. Upon detecting biases, users can refine GenAI systems by infusing their datasets or system prompts with more contextual information. Subsequently,

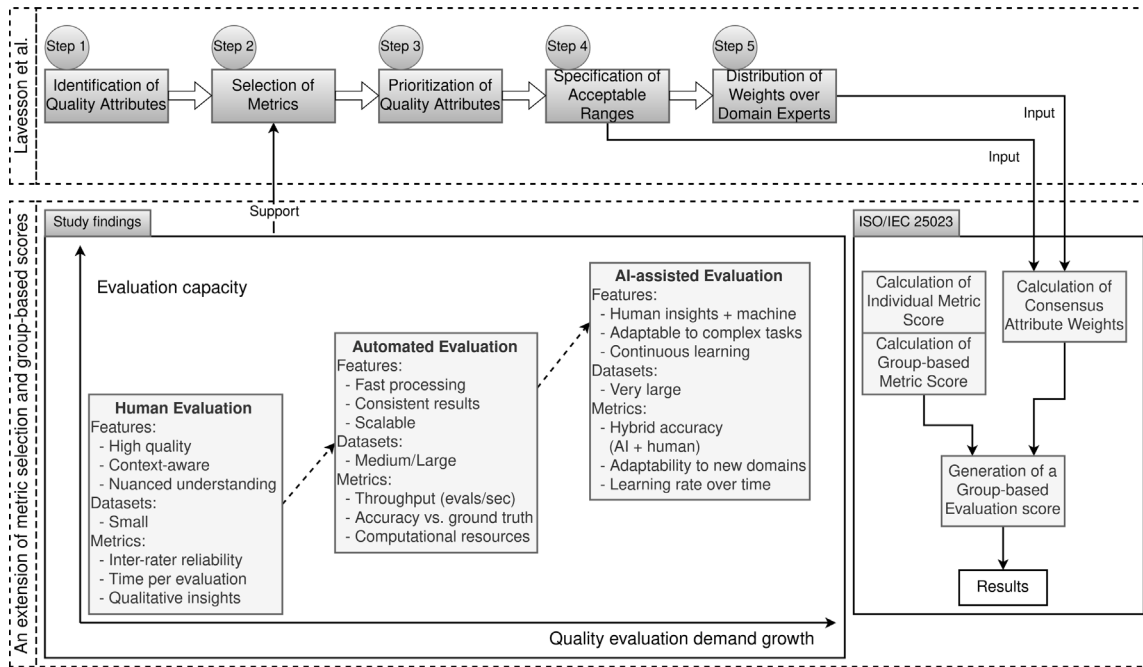


Fig. 9. A framework for quality evaluation of GenAI systems.

recalculating the scores will indicate if the adjustments have mitigated the bias.

Key considerations for metric selection: Given the limitations reported in RQ2 (Sections 4.2.2 and 4.2.3), metric selection should be guided by:

1. Relevance to use cases: Choose metrics that align with the specific quality characteristics essential for the GenAI application (e.g., BERTScore for coherence, FActScore for factual accuracy).
2. Dataset quality and availability: Ensure datasets are representative and free from biases to maintain evaluation integrity.
3. Combined evaluation methods: Integrate automated metrics with human or AI-assisted evaluations for a more thorough assessment.
4. Scalability and resource constraints: Consider the computational resources required, especially for large datasets or complex models.

5. A suggested framework for evaluating generative AI systems

Our study results show that a GenAI system evaluation may require assessing a set of risky output dimensions to continuously measure and manage quality characteristics in an industrial application. Such a framework for this purpose is needed based on our findings in this study.

Therefore, as shown in Fig. 9, we suggest a framework that supports practitioners with multi-methods, metrics, and processes for quality attribute evaluation of GenAI systems. This framework is inspired by the research work of Lavesson et al. [16] on an application-oriented-validation-evaluation method. Their method contains five sequential and general steps to address the evaluation of AI components based on quality requirements prior to the inclusion of the components in software systems/products. However, detailed information about existing evaluation metrics or methods is missing. We extend their method by filling in the missed information.

As shown in Fig. 9, the proposed framework consists of five steps. Steps 1 to 5 are initially from the study of Lavesson et al. [16].

Step 1: Identification of quality characteristics. Product designers or architects should provide an initial list of quality characteristics. When these are unclear, the ISO/IEC 25023 standards can assist in defining them based on our experience.

Step 2: Selection of metrics. Our study findings show that metrics are associated with evaluation methods or approaches. In practice, the choice of evaluation method depends on the specific industrial requirements in Step 1. Human evaluation is an initial approach for assessing GenAI system outputs on a limited set of user queries and small datasets. As the scale of evaluation increases or the need for continuous quality monitoring grows, automated evaluation becomes a more viable option. AI-assisted evaluation offers a compelling solution for mission-critical applications (e.g., healthcare) requiring strict accuracy with factual sources. It enhances automated evaluations with deeper contextual insights and better alignment with human expectations.

Moreover, the progression from human to automated to AI-assisted evaluation methods showcases a steadily increasing capacity in assessing GenAI systems. Here, capacity refers to the evaluation method's ability to capture context, semantic nuance, and domain-specific criteria, not its throughput or scalability. While automated methods generally scale better with input volume, they may lack the interpretive depth required in complex or safety-critical domains. Each method has its place in the evaluation toolkit, and the choice among them can be guided by the process defined in Lavesson et al. [16].

Step 3: Prioritization of quality requirements. Consider a real-world application of supervised learning where multiple quality characteristics are recognized. It is unlikely that these attributes hold equal significance. According to Lavesson et al. individual metric weights provided by domain experts worked in this step. As mentioned, each identified metric is associated with an ISO quality attribute, which can be treated as quality attribute weights.

Step 4: Specification of acceptable ranges. We inherit this step from the work of Lavesson et al. [16]. It means that each domain expert — i ($i = 1, 2, 3, \dots, k$) is allowed to associate with each selected metric — j ($j = 1, 2, 3, \dots, n$) an acceptable range — r_{ij} ($r_{ij} = [b_{ij}^l, b_{ij}^u]$), where the former denotes the least desired acceptable score, and latter denotes the desired score.

Table 14
Maturity measures [22] in ISO/IEC 25023.

ID	Name	Description	Measurement function
RMa-2-G	Mean time between failure (MTBF)	What is the MTBF during the system/software operation?	$X = A/B$, where A=number of failures detected during observation time, B=duration of observation.

Step 5: Distribution of weights over domain experts. Domain experts are permitted to rate their peers, thus indicating the extent of influence they are willing to accept from the group. The detailed decision process is based on the study by Lavesson et al. [16].

Furthermore, to streamline the definition and calculation of metric formulas, we utilize ISO/IEC 25023, which provides a clear definition and formula for quality measures. Practitioners may follow the established ISO calculations or tailor these calculations to obtain their evaluation scores. An example of maturity measures within the ISO standards is described in Table 14.

In short, the steps in this proposed framework have been validated by several empirical case studies, such as [53–55]. Consequently, this framework serves as a supportive guideline to assist engineers when they have a need to introduce quality assessment and monitoring of GenAI systems within the industry.

6. Validity threats

Following Ampatzoglou et al. [56], we analyze threats using their framework for literature review studies.

6.1. Study selection validity

A threat is potential selection bias in the snowballing start-set (Table 1) due to reliance on recent literature reviews, which might exclude older papers. Forward/backward snowballing could exclude some related studies that are not indexed in bibliographic databases. To mitigate this, we combined snowballing with review of the ArXiv repository to capture grey literature and documented inclusion/exclusion criteria transparently (Table 2).

6.2. Data validity

A threat is a subjectivity in mapping metrics to ISO characteristics (e.g., METEOR to “Appropriateness Recognizability”). We reduced this via a six-step protocol (Table 7) with three authors cross-validating mappings. Another threat is inconsistency in rigor/relevance scoring during paper filtering (Section 3.3.2). We calculated Fleiss’ $\kappa = 0.57$ to quantify inter-rater reliability and resolved discrepancies through group discussions. The oversimplification of evaluation methods (human/automated/AI-assisted) was mitigated by grounding classifications in inductive coding (Fig. 2).

6.3. Research validity

A threat is domain specificity: findings for text-based GenAI systems (e.g., document generation) may not generalize to multimedia domains like image/video generation. We explicitly scoped the study to text-based GenAI applications (Section 1). Reliance on ISO/IEC 25023, which lacks GenAI-specific traits (e.g., non-determinism), was mitigated by proposing framework extensions (Section 5). Variance in metric implementation (e.g., Bleurt) was mitigated by providing implementation repositories and examples (Tables 12). Small sample rigor/relevance assessments (3/69 papers) were offset by transparent reporting and inter-rater agreement analysis.

7. Discussions

7.1. Datasets required for automated metrics

The result of automated evaluation metrics is influenced by the quality and relevance of the datasets used. High-quality, domain-specific datasets enhance the accuracy of metrics like BERTScore, which relies on reference data for assessment. However, challenges arise when datasets are outdated, biased, or incomplete, potentially leading to misleading results. Metrics like SARI, for example, may penalize valid semantic variations that differ from reference texts, overlooking the richness of meaning.

This dependency on high-quality datasets is problematic in domains like Healthcare, where such datasets are not easy to acquire. Although automated metrics offer scalable assessments, their limitations highlight the need to complement them with AI-assisted techniques to gauge the qualities of datasets, for example, AI synthetic data.

7.2. Cost versus quality

Implementing evaluation metrics can be resource-intensive in the industry. Organizations should balance the cost of these metrics with the potential benefits while using them. Advanced metrics like MoverScore and COMET provide deeper insights but come with increased computational demands. As Zhao et al. [21] suggest, conducting cost-benefit analyses is crucial for determining which metrics align best with an organization’s budget and quality objectives.

Lavesson et al. [16] further propose an analytic hierarchy process that engages domain experts in prioritizing quality characteristics through pairwise comparisons. This pragmatic approach aids organizations in optimizing metric selection according to their specific needs and constraints.

7.3. Selecting appropriate metrics to evaluate system output quality

The selection of an appropriate set of metrics is crucial for evaluating GenAI system output quality. Lavesson et al. [16] emphasize the importance of involving domain experts in this process, offering an evaluation method designed for industrial applications. A combination of human, automated, and AI-assisted evaluation methods can yield a more in-depth assessment, covering a wider range of quality characteristics.

Eddine et al. [28] support this multi-method approach, advocating for a combination of metrics to capture diverse aspects of output quality. This suggests that practitioners should not rely on a single evaluation method but instead employ a blended approach to achieve more impact and nuanced assessments.

7.4. Risky outputs exist in GenAI industrial applications

Despite the potential of using metrics to mitigate risks, a significant challenge hindering GenAI adoption in the industry is its unpredictable nature. Minor changes in input prompts, such as punctuation or abbreviations, can lead to vastly different outputs, which can negatively affect user experience and system reliability.

Although dynamic data sampling could help address this unpredictability, current metrics typically rely on static datasets, as indicated by our study findings. Therefore, further research is needed to develop

Table A.15
Selected papers.

ID	Author(s)	Paper title	Year
[S1]	Kynkäänniemi et al.	Improved Precision and Recall Metric for Assessing Generative Models	2019
[S2]	Fabbri et al.	SummEval: Re-evaluating Summarization Evaluation	2021
[S3]	Huang et al.	AgentCoder: Multi-Agent-based Code Generation with Iterative Testing and Optimization	2020
[S4]	Ghazarian et al.	Better automatic evaluation of open-domain dialogue systems with contextualized embeddings	2019
[S5]	Gruver et al.	Large language models are zero-shot time series forecasters	2024
[S6]	Mathur et al.	Tangled up in BLEU: Reevaluating the evaluation of automatic machine translation evaluation metrics	2020
[S7]	Chen et al.	Evaluating Large Language Models Trained on Code	2021
[S8]	Rei et al.	COMET: A neural framework for MT evaluation	2020
[S9]	Eddine et al.	FrugalScore: Learning Cheaper, Lighter and Faster Evaluation Metrics for Automatic Text Generation	2021
[S10]	Honovich et al.	TRUE: Re-evaluating Factual Consistency Evaluation	2022
[S11]	Sellam et al.	BLEURT: Learning Robust Metrics for Text Generation	2020
[S12]	Xu et al.	Optimizing statistical machine translation for text simplification	2016
[S13]	Galley et al.	deltaBLEU: A Discriminative Metric for Generation Tasks with Intrinsically Diverse Targets	2015
[S14]	Nguyen et al.	The effectiveness of feature attribution methods and its correlation with automatic evaluation scores	2024
[S15]	Mastropaolo et al.	Evaluating Code Summarization Techniques: A New Metric and an Empirical Characterization	2024
[S16]	Nallapati et al.	Abstractive text summarization using sequence-to-sequence rnns and beyond	2024
[S17]	Bakir et al.	Developing and Evaluating a Model-Based Metric for Legal Question Answering Systems	2023
[S18]	Pillutla et al.	MAUVE: Measuring the Gap Between Neural Text and Human Text using Divergence Frontiers	2021
[S19]	Kamalloo et al.	Evaluating open-domain question answering in the era of large language models	2023
[S20]	Zhang et al.	BERTScore: Evaluating Text Generation with BERT	2019
[S21]	Mathur et al.	Putting evaluation in context: Contextual embeddings improve machine translation evaluation	2019
[S22]	Chowdhery et al.	Palm: Scaling language modeling with pathways	2023
[S23]	Wu et al.	A comparative study of open-source large language models, gpt-4 and claude 2: Multiple-choice test taking in nephrology	2023
[S24]	Kang et al.	Identifying Inaccurate Descriptions in LLM-generated Code Comments via Test Execution	2024
[S25]	Stanojevic et al.	Beer: Better evaluation as ranking	2018
[S26]	Palotti et al.	TrecTools: an open-source Python library for Information Retrieval practitioners involved in TREC-like campaigns	2019
[S27]	Lardilleux et al.	CHARCUT: Human-targeted character-based MT evaluation with loose differences	2017
[S28]	Lo et al.	YiSi-a unified semantic MT quality evaluation and estimation metric for languages with different levels of available resources	2019
[S29]	Florian et al.	Exploring Precision and Recall to assess the quality and diversity of LLMs	2024
[S30]	Zhao et al.	MoverScore: Text Generation Evaluating with Contextualized Embeddings and Earth Mover Distance	2019
[S31]	Krscinski et al.	Evaluating the Factual Consistency of Abstractive Text Summarization	2019
[S32]	Wang et al.	Asking and Answering Questions to Evaluate the Factual Consistency of Summaries	2020
[S33]	Le et al.	Coder1: Mastering code generation through pretrained models and deep reinforcement learning	2022
[S34]	Graham et al.	Accurate Evaluation of Segment-level Machine Translation Metrics	2015
[S35]	Apidianaki et al.	METEOR-WSD: improved sense matching in MT evaluation	2015
[S36]	Toutanova et al.	A dataset and evaluation metrics for abstractive compression of sentences and short paragraphs	2016
[S37]	Johnson et al.	Assessing the Accuracy and Reliability of AI-Generated Medical Responses: An Evaluation of the Chat-GPT Model	2023
[S38]	Vaithilingam et al.	Expectation vs. Experience: Evaluating the Usability of Code Generation Tools Powered by Large Language Models	2022
[S39]	Min et al.	FActScore: Fine-grained atomic evaluation of factual precision in long form text generation	2023
[S40]	Chia et al.	INSTRUCTEVAL: Towards holistic evaluation of instruction-tuned large language models	2023
[S41]	Sharma et al.	An Empirical Study of Unsupervised Evaluation Metrics for Dialogue Response Generation	2017
[S42]	Moramarco et al.	Human evaluation and correlation with automatic metrics in consultation note generation	2022
[S43]	Ren et al.	Codebleu: a method for automatic evaluation of code synthesis	2020

more dynamic evaluation methods that can accommodate the inherent variability of GenAI-generated outputs, ensuring they meet industry quality standards.

7.5. The potential for undiscovered risky outputs and the need for new metrics

As GenAI technology continues to evolve, new forms of risky outputs are likely to emerge from novel applications, unforeseen interactions with other systems, or evolving ethical considerations. For example, risky outputs may occur when interacting with other GenAI systems or complex infrastructures.

The possibility of such undiscovered risks highlights the need for new metrics and evaluation methods. Future research should focus on developing metrics that assess not only GenAI system outputs but also their impacts, involving experts from fields such as ethics and cybersecurity.

7.6. Limitations of snowballing with a small start-set

While our snowballing enabled a focused synthesis of literature on metrics for GenAI systems, it introduces limitations compared to traditional systematic literature reviews. Specifically, the reliance on 12 SLRs as a start-set — selected for their thematic relevance and citation impact — may narrow the selection of the start-set. Jalali

and Wohlin's comparative analysis [44] on research topic 'Agile practices in global software engineering' underscores that snowballing and database searches result in different paper sets, though converging on similar conclusions. However, no direct comparative analysis exists between snowballing and SLR outcomes in the domain of GenAI metrics, limiting our ability to assess potential coverage gaps. Future studies could evaluate the differences between these methods empirically in synthesizing GenAI metrics.

8. Conclusions

This study focused on exploring and identifying metrics for evaluating the quality of outputs generated by GenAI systems. Through a snowballing literature review, we identified 28 metrics mapped to four ISO/IEC 25023 quality characteristics: usability, reliability, functional suitability, and maintainability. Our findings suggest that these metrics provide valuable insights for the quality evaluation of GenAI systems, although our conclusions are derived from qualitative synthesis rather than quantitative measurements.

In our analysis of human, automated, and AI-assisted evaluation methods, we observed that a combined approach might offer a broader evaluation based on recurring themes across the reviewed studies. Specifically, our review indicates that: (I) Human evaluation, while context-aware, may be influenced by individual biases; (II) Automated methods deliver scalability but depend heavily on the quality of the underlying datasets; (III) AI-assisted methods can enhance adaptability,

but they may introduce additional complexity in terms of deployment and computational requirements. We emphasize that the methods' robustness and deployment complexity emerged from qualitative interpretations of the literature and were not measured directly.

Building on these insights, we proposed a framework, guided by Lavesson et al.'s method, to assist industrial engineers in selecting appropriate quality evaluation methods for GenAI system outputs according to their quality requirements.

In conclusion, this study offers primarily practitioners, but also researchers, a set of metrics, evaluation methods, and processes to improve the quality assessment of GenAI system outputs. Continuous refinement and adaptation of these evaluation methods will be essential as GenAI applications become increasingly integrated into industrial contexts.

CRedit authorship contribution statement

Liang Yu: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Emil Alégroth:** Writing – review & editing, Validation, Supervision, Formal analysis. **Panagiota Chatzipetrou:** Writing – review & editing, Supervision, Formal analysis. **Tony Gorschek:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Selected papers after snowballing literature review

See Table A.15.

Data availability

Data will be made available on request.

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